

9. Compare Linear and Polynomial Regression using Python.

CODE:

```
[9]
✓ Os
from sklearn.datasets import load_diabetes
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.preprocessing import PolynomialFeatures
from sklearn.metrics import r2_score

# Load dataset
data = load_diabetes()
X = data.data
y = data.target

# Split dataset
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=1
)

# Linear Regression
linear = LinearRegression()
linear.fit(X_train, y_train)
linear_pred = linear.predict(X_test)

# Polynomial Regression (Degree = 2)
poly = PolynomialFeatures(degree=2)
X_train_poly = poly.fit_transform(X_train)
X_test_poly = poly.transform(X_test)
polynomial = LinearRegression()
polynomial.fit(X_train_poly, y_train)
poly_pred = polynomial.predict(X_test_poly)

# Display Results
print("Linear Regression R2 Score :", r2_score(y_test, linear_pred))
print("Polynomial Regression R2 Score :", r2_score(y_test, poly_pred))
```

OUTPUT:

```
... Linear Regression R2 Score : 0.43845439143447806
Polynomial Regression R2 Score : 0.21244369187731138
```